AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 30 · STUDENT LESSON

Dataset shift in neural models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Count the groups. Weight the rates. Check the evidence.

Explain how overall accuracy can change with an unchanged model and unchanged observed group rates.

Keep one habit: Complete counts → paired bar chart → weighted calculation → conditional scenario → separate picture → investigation plan → independent audit.

MAIN LESSON · SECTION 1

Can the overall score change when group rates stay the same?

> Track group counts as well as accuracy.

Learning goals

I can explain how a changed **input** mix can change overall accuracy.

I can calculate group and overall results with the correct denominators.

I can plan a shift investigation while preserving independent evaluation.

Retrieve monitoring and evaluation

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Can a high confidence score establish that a new **input** resembles the development data or that its **prediction** is correct?

> **My starting explanation**

End goal

By the end, I can explain **dataset shift** in neural models using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain how a changed **input** mix can change overall accuracy

Calculate group and overall results with the correct denominators

Plan a shift investigation while preserving independent evaluation

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Hold the model fixed and inspect who appears in the data

> Every audit case belongs to one declared group.

Dataset shift and controlled scope

A neural image classifier developed using mostly clear pictures may later receive mostly dim pictures. Its learned weights can stay unchanged while its overall results change.

Dataset shift concerns changes in the data encountered, including **input**s, label frequencies or the relationship between **input**s and labels.

A shift does not always reduce accuracy, and an accuracy change alone does not identify its cause.

Use the fictional audit below. Clear and dim are disjoint groups defined before the audit, and every case belongs to exactly one group. The same model checkpoint, pre**process**ing and **prediction** threshold are used throughout.

Counts are constructed teaching data, not results from a deployed model. Both cohorts have checked reference labels.

Complete group counts and correct counts

Cohort	Group	Cases	Correct
Development audit	Clear	80	72
Development audit	Dim	20	10
New-use audit	Clear	20	18
New-use audit	Dim	80	40

Compare the observed within-group rates

Clear accuracy is 72 / 80 = 90 percent in development and 18 / 20 = 90 percent in new use. Dim accuracy is 10 / 20 = 50 percent and 40 / 80 = 50 percent.

The observed group rates are unchanged, but the group proportions are different.

Name supplied and missing facts

List the given facts and one fact that is not supplied.

Add counts first, then divide by the total

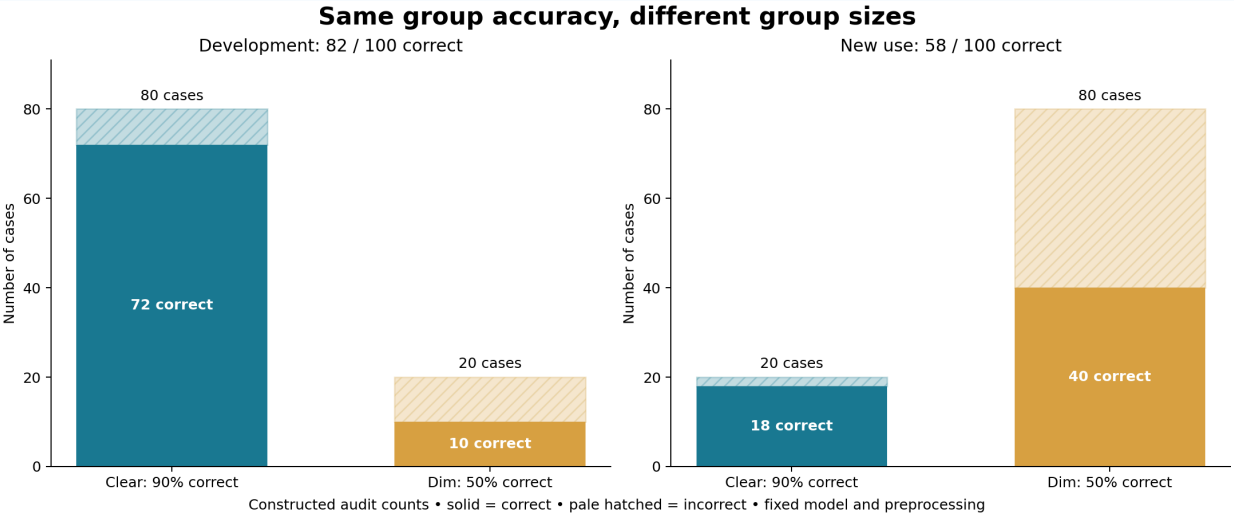
> An average of percentages needs the group-size weights.

Compute both overall rates and the percentage-point drop

Development accuracy is (72 + 10) / (80 + 20) = 82 percent. New-use accuracy is (18 + 40) / (20 + 80) = 58 percent. The drop is 24 percentage points.

Taking the unweighted mean of 90 and 50 gives 70 percent, which is not either cohort's overall accuracy because neither cohort has equal group sizes.

See the changing group sizes and counts



Original runnable arithmetic

```
def accuracy(groups):
    if not groups:
        raise ValueError("Need at least one group")
    total = correct = 0
    for n, k in groups:
        if n <= 0 or not 0 <= k <= n:
            raise ValueError("Invalid group counts")
```

```
        total += n
        correct += k
    return correct / total

development = [(80, 72), (20, 10)]
new_use = [(20, 18), (80, 40)]
old = accuracy(development)
new = accuracy(new_use)
print(round(100 * old, 1))  # 82.0
print(round(100 * new, 1))  # 58.0
print(round(100 * (old - new), 1))  # 24.0
```

Disjoint groups, valid counts and zero-case limits

Each pair is (cases, correct), and the function assumes integer counts for non-overlapping groups. It rejects empty **input**s, groups with no cases and impossible correct counts.

A group with no observed cases has an undefined empirical accuracy, not zero accuracy. Do not count the same example in two groups when forming an overall total.

Check using the weighted-rate formula

Equivalently, weight each group accuracy by its share of cases: development is 0.80 times 0.90 + 0.20 times 0.50 = 0.82; new use is 0.20 times 0.90 + 0.80 times 0.50 = 0.58.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

Predict a new mixture under an explicit assumption

> Keep the assumed group rates separate from measured outcomes.

Current task and response

- Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed picture

- Change: New-use mix returns to40 bright and 10 dim with the same slice rates.

▼ Check after predicting

Prediction: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice’s conditional performance stays fixed.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

Distinguish an observed count from a conditional scenario

> Both examples keep within-group rates fixed while changing proportions.

Main mixture explanation and limits

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

The arithmetic explains this constructed aggregate difference. It does not prove that dim lighting caused each real error or that these group rates will stay fixed in future samples.

Report group counts and uncertainty about generalisation. Review specific errors and reference labels as in Week 29.

Try an equal-sized main cohort

If a new cohort has 50 clear and 50 dim cases, and exactly 90 percent and 50 percent are correct respectively, how many correct cases and what overall accuracy result? State the assumption behind this calculation.

Separate picture inputs and denominators

Separate within-setting performance from changing setting proportions.

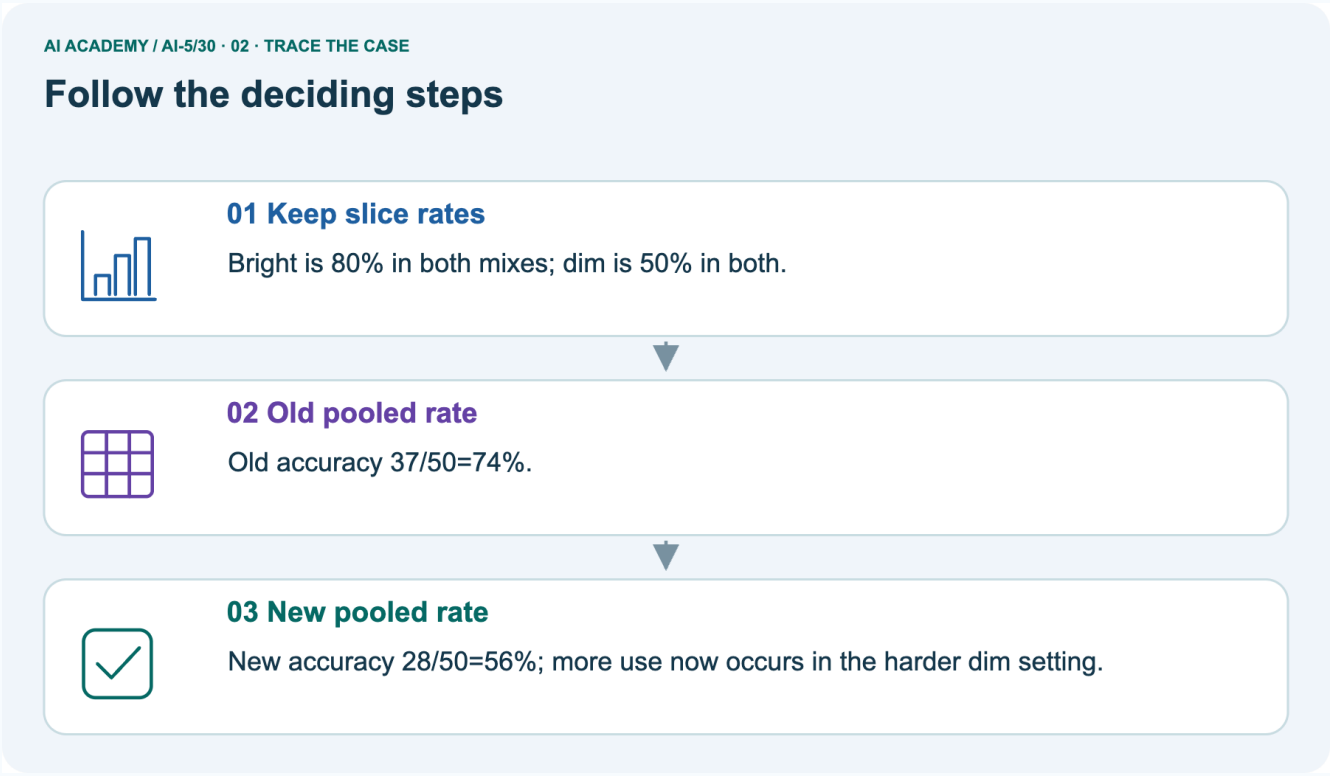
Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate 50-case example: bright accuracy is 80%, not the main clear-group rate of 90%. Old picture counts 40 bright/32 correct plus 10 dim/5 correct give 37/50 = 74%. New counts 10 bright/8 correct plus 40 dim/20 correct give 28/50 = 56%. Returning to the old proportions under the same rates returns 74%.

This isolates a mixture calculation; it does not establish future rates or causal lighting effects.

Trace the pictured weighted result



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep slice rates	Bright is 80% in both mixes; dim is 50% in both.
2. Old pooled rate	Old accuracy 37/50=74%.
3. New pooled rate	New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

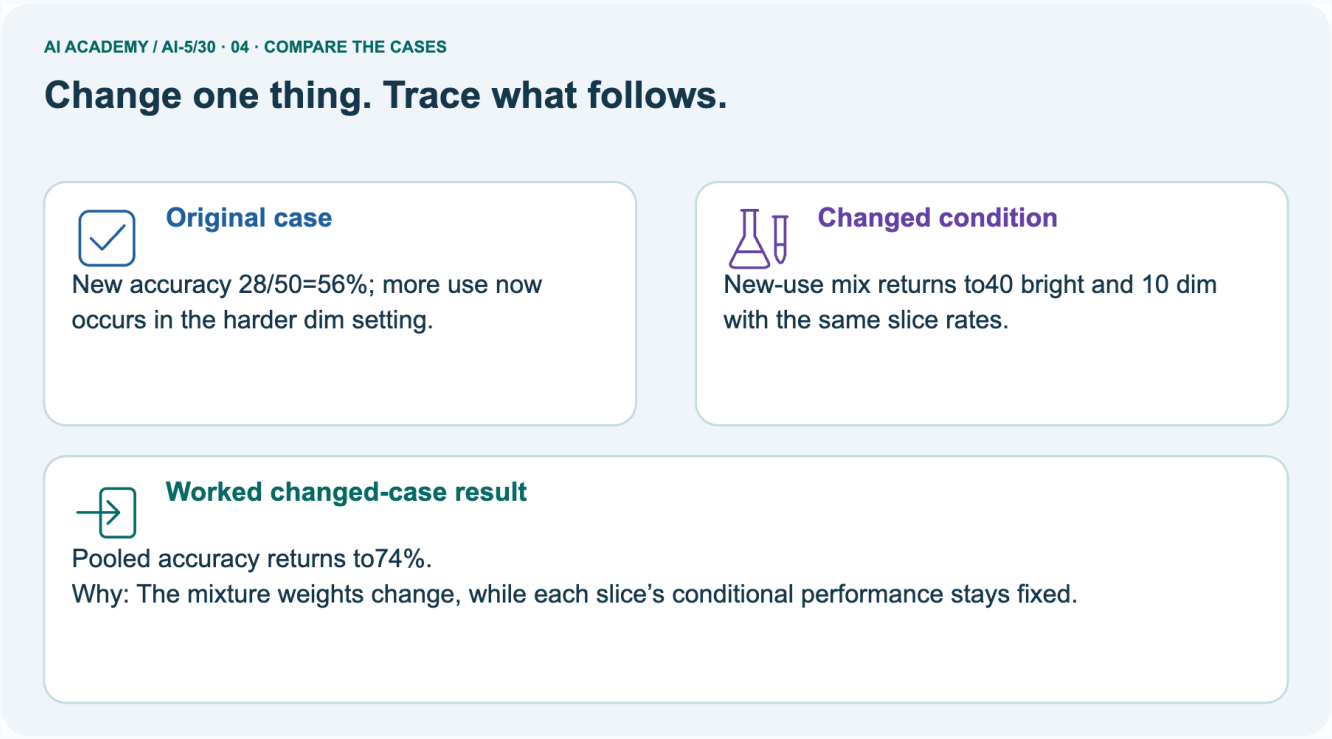
A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Predict restoring the pictured proportions

- Change one thing: New-use mix returns to40 bright and 10 dim with the same slice rates. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the unchanged group rate and remaining limits

- Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

- Read the labelled picture: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

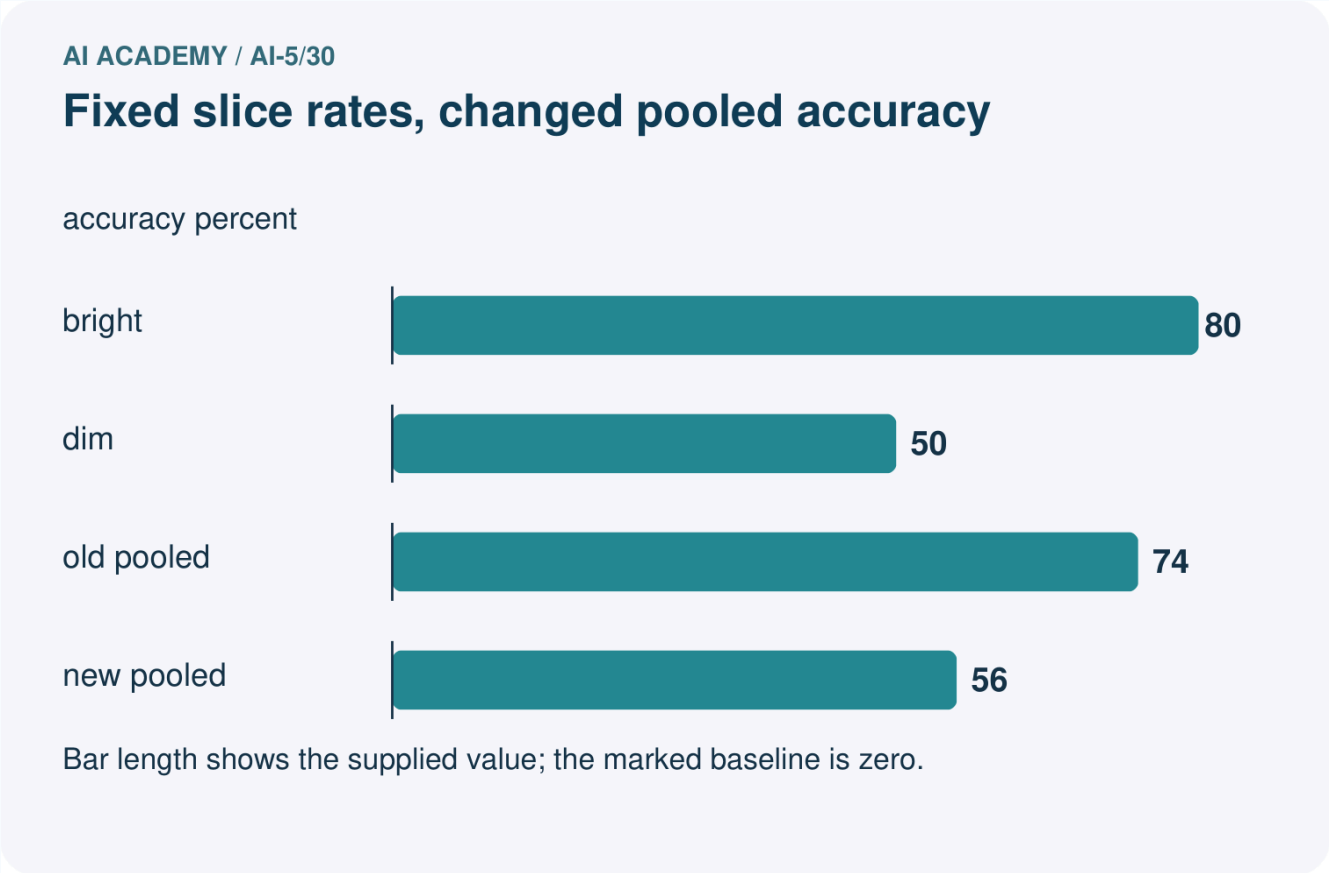
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dataset shift in neural models

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/30

Dataset shift in neural models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep slice rates

Bright is 80% in both mixes; dim is 50% in both.



2. Old pooled rate

Old accuracy 37/50=74%.



3. New pooled rate

New accuracy 28/50=56%; more use now occurs in the harder dim setting.

CHANGE ONE THING

New-use mix returns to40 bright and 10 dim with the same slice rates.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Check the pipeline and labels before choosing a remedy

> Monitoring alone cannot reveal every kind of shift.

Version checks, unavailable accuracy and controlled investigation

Start with model and **input**-pipeline version checks. A changed resize operation, channel order or unit conversion is an implementation change, not **evidence** that the **input** population itself shifted.

Compare image conditions, missing values, source groups and label definitions.

A neural network may have learned a shortcut tied to a background or capture condition; that is a hypothesis to test, not a conclusion from a histogram.

Input monitoring without new reference labels can reveal a changed **group mix**. It cannot calculate current accuracy or prove that the relationship between **input**s and labels stayed the same.

Conversely, stable **input** summaries can miss a changed labelling policy. Obtain a representative, correctly labelled audit sample and document unresolved label quality.

For a controlled lighting study, keep scene identity and the label fixed while varying lighting. Keep related images together in one split.

If new data guide re**training** or threshold choices, they are development data; reserve separate source groups or a later appropriate time period for final evaluation. More data alone do not guarantee that a shortcut is removed.

Evidence-to-action plan

Evidence or check	Permitted conclusion / next action
More dim inputs, no new checked labels	Changed input mix observed; current accuracy unavailable
Same input histogram, changed label policy	Relationship can change without an obvious input-summary change

Evidence or check	Permitted conclusion / next action
Different resize, channel order or units	Check implementation versions before attributing results to population shift
New paired lighting comparisons	Keep scene identity and label fixed; keep related source images in one split
Retraining uses these new errors	Those cases become development evidence; reserve new appropriate final cases
Original versus revised model comparison	Run both on identical reserved cases; retain baseline, subgroup counts and workload
No cases in a declared group	Empirical group accuracy unavailable; do not record zero accuracy

Try the no-label monitoring question

- New-use **monitoring** shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise group rates, group shares and weighted totals

> Write the denominator beside every rate.

Current task and conditional equal mixture

- From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

> **If a new cohort has 50 clear and 50 dim cases, and exactly 90 percent and 50 percent are correct respectively, how many correct cases and what overall accuracy result? State the assumption behind this calculation.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Dataset shift** in neural models". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual weighting misconception

- Use the worked case for **Dataset shift** in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured unchanged-rate question

- Did the supplied model’s dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve independent P/Q audit

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Use the right denominator

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete weighted-count audit

> Validate counts, preserve denominators and distinguish observations from scenarios.

Current overall-count investigation

Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Complete runnable audit and separate picture

```
"""Constructed disjoint-group counts; unchanged model and preprocessing."""
def audit(groups):
    if not groups:
        raise ValueError('Need at least one disjoint group')
    total = correct = 0
    ledger = []
    for name, n, k in groups:
        if type(n) is not int or type(k) is not int or n <= 0 or not 0 <= k <= n:
            raise ValueError('Need positive integer cases and valid integer correct
count')
        total += n
        correct += k
        ledger.append((name,n,k,k/n))
    weighted = sum(n/total*rate for _,n,_,rate in ledger)
    return {'cases':total,'correct':correct,'errors':total-correct,
            'accuracy':correct/total,'weighted_accuracy':weighted,'groups':ledger}
development = [('clear',80,72),('dim',20,10)]
new_use = [('clear',20,18),('dim',80,40)]
old,new = audit(development),audit(new_use)
assert old['correct']==82 and new['correct']==58
assert old['errors']==18 and new['errors']==42
assert [r[3] for r in old['groups']]==[r[3] for r in new['groups']]==[.9,.5]
assert abs(old['accuracy']-old['weighted_accuracy'])<1e-12
assert abs(new['accuracy']-new['weighted_accuracy'])<1e-12
print('Development:',old)
print('New use:',new)
print('Drop in percentage points:',round(100*(old['accuracy']-new['accuracy']),8))
# Conditional scenario: assume these exact within-group rates in the new cohort.
```

```
equal_mix=audit([('clear',50,45),('dim',50,25)])
assert equal_mix['accuracy']==.7
print('Conditional equal mix:',equal_mix)
# Separate picture uses bright accuracy .8, not the main clear accuracy .9.
picture_old=audit([('bright',40,32),('dim',10,5)])
picture_new=audit([('bright',10,8),('dim',40,20)])
assert picture_old['accuracy']==.74 and picture_new['accuracy']==.56
print('Separate picture:',picture_old['accuracy'],picture_new['accuracy'])
# Without checked reference labels, no current accuracy can be computed.
# Group names alone cannot detect overlapping source records; audit source IDs separately.
```

Expected outputs and invalid-input probes

1. Predict both correct totals and error totals before running or tracing. Development: 82 correct and 18 errors of 100; new use: 58 correct and 42 errors of 100. The fall is 24 percentage points. Confirm weighted and pooled calculations agree. Check empty input, a zero-case group and impossible or noninteger counts: the complete audit rejects them.

   Group names alone cannot detect duplicated source records, so source-group membership needs a separate check. The equal-mix result assumes the stated rates hold exactly; no new model evaluation or future reliability is claimed.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Reference: earlier arithmetic helper

```
def accuracy(groups):
    if not groups:
        raise ValueError("Need at least one group")
    total = correct = 0
    for n, k in groups:
        if n <= 0 or not 0 <= k <= n:
            raise ValueError("Invalid group counts")
        total += n
        correct += k
    return correct / total

development = [(80, 72), (20, 10)]
new_use = [(20, 18), (80, 40)]
old = accuracy(development)
new = accuracy(new_use)
print(round(100 * old, 1))  # 82.0
print(round(100 * new, 1))  # 58.0
print(round(100 * (old - new), 1))  # 24.0
```

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compute a fresh P/Q mixture

> Use the new counts and give a percentage-point change.

Complete independent audit and scenario

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Analyse a new mixture independently

Attempt this without hints

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the reference cohort and model versions in the project

> Save the original model for a comparable later evaluation.

Project artifact and evidence

Add a labelled **dataset shift** in neural models **evidence** sheet to Image-pattern Network Investigation. Show how you can explain how a changed **input** mix can change overall accuracy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the mixture change without overclaiming its cause

> A pooled score needs group counts, checked labels and a comparison period.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain how a changed **input** mix can change overall accuracy

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Dataset shift	A change in the distribution of inputs, labels or their relationship between settings
Subgroup	A declared subset of examples sharing a relevant condition
Group mix	The proportions of examples belonging to different groups
Monitoring	Repeatedly checking recorded inputs, outputs and available outcomes against a reference

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

- New-use **monitoring** shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

- A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent **evidence** and propose a comparison that includes the original model.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the independent P/Q audit for the first individual attempt.

Current explanation and response

- Possible responses include fixing a confirmed pipeline bug, gathering missing **training** coverage, changing a representation or routing some cases for review. Choose a response based on **evidence**.

  Compare the original and revised model on identical reserved cases, report both group and overall results, and measure review workload if a review policy is used. Preserve the original checkpoint for comparison.

- Record the reference period, group definitions, model and pipeline versions, counts, available labels, decisions and follow-up outcomes. Avoid silently replacing the baseline or the evaluation set.

  Next week measures efficiency and resource use alongside model quality.

- A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent **evidence** and propose a comparison that includes the original model.

> **A team sees more dim images but has no new reference labels. It announces accuracy has fallen, retrains, and claims success using the same cases that guided its changes. Identify what is known, what is missing and how to repair the evaluation.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Separate within-setting performance from changing setting proportions.

By the end, I can	How I will show it
Explain how a changed input mix can change overall accuracy	Explain the deciding step, show a trace or test, and state one limit.
Calculate group and overall results with the correct denominators	Explain the deciding step, show a trace or test, and state one limit.
Plan a shift investigation while preserving independent evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate within-setting performance from changing setting proportions.
- Which labelled step matches this detail? New accuracy 28/50=56%; more use now occurs in the harder dim setting.
- 1. New pooled rate
- 2. Keep slice rates
- 3. Old pooled rate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: New pooled rate New accuracy 28/50=56%; more use now occurs in the harder dim setting.

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Changed-case question and optional hint

- Changed-case practice: New-use mix returns to40 bright and 10 dim with the same slice rates. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Pooled accuracy returns to74%.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Pooled accuracy returns to74%. The mixture weights change, while each slice's conditional performance stays fixed.

Repair and retain the record

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

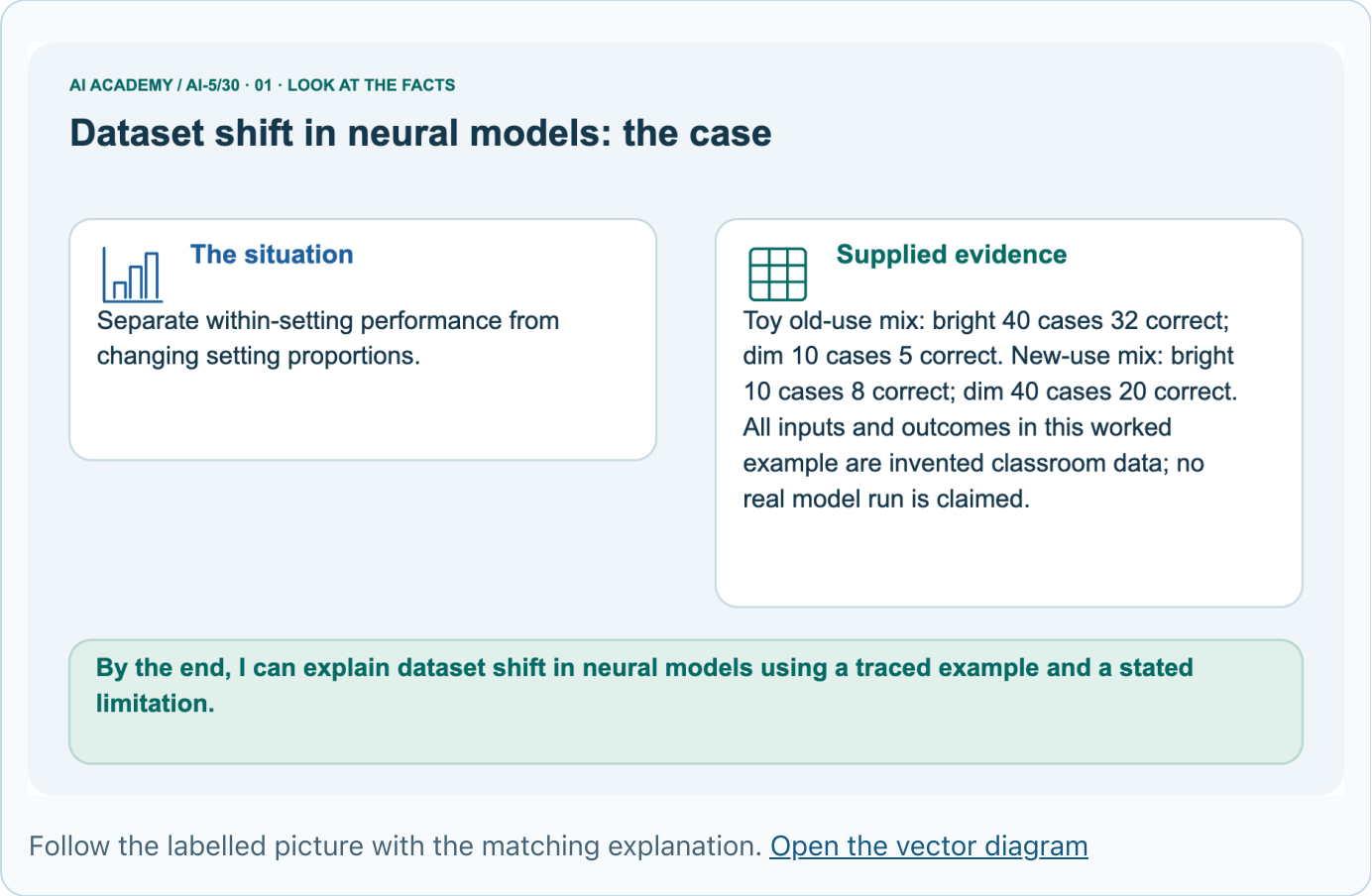
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Dataset shift in neural models: the case



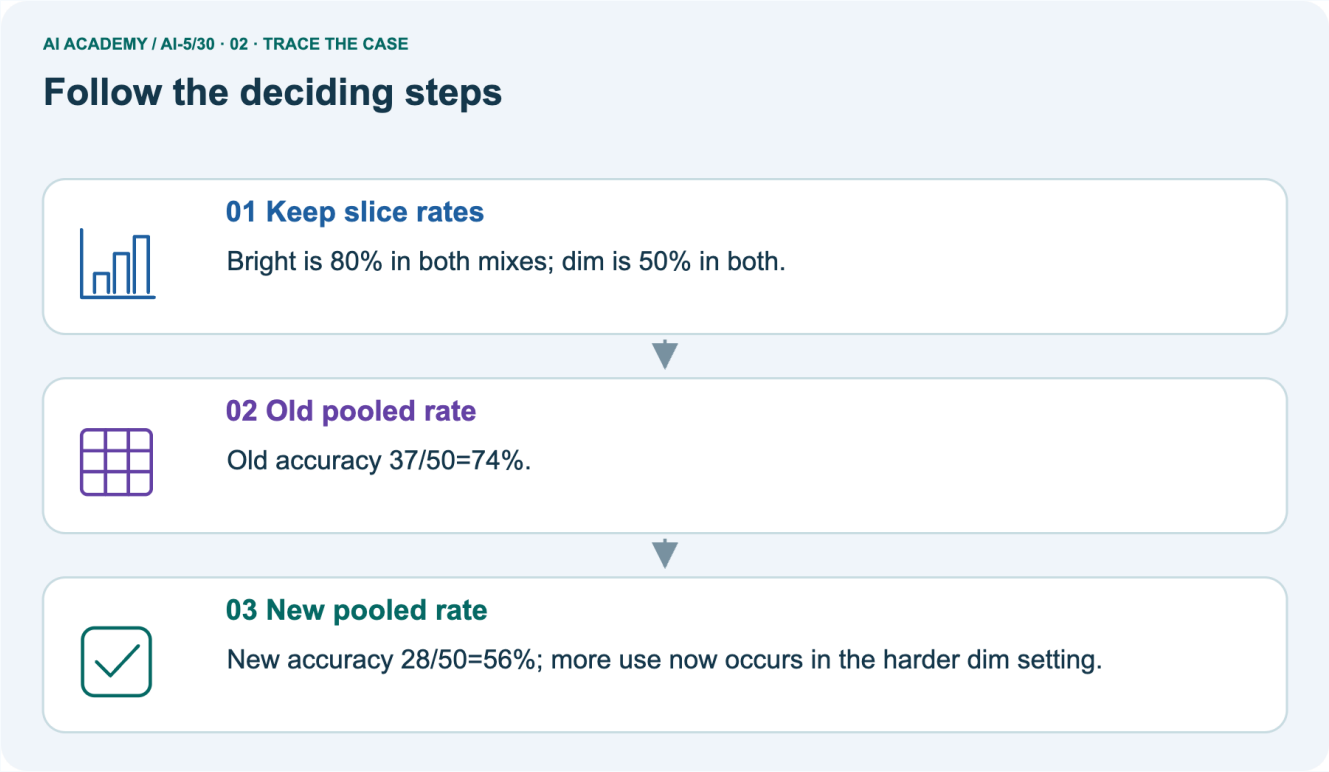
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate within-setting performance from changing setting proportions. Supplied facts: Toy old-use mix: bright 40 cases 32 correct; dim 10 cases 5 correct. New-use mix: bright 10 cases 8 correct; dim 40 cases 20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

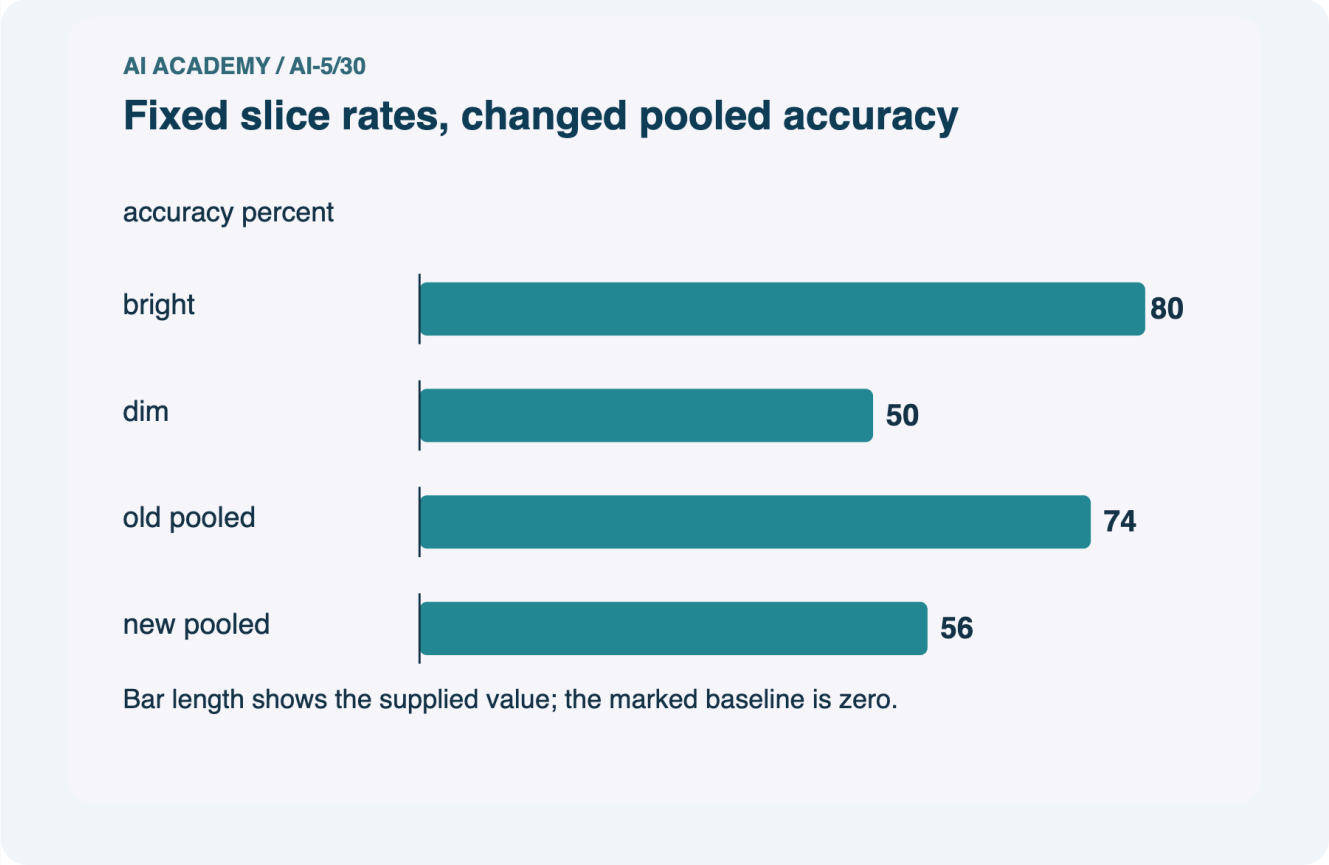


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep slice rates: Bright is **80%** in both mixes; dim is 50% in both. → Old pooled rate: Old accuracy 37/50=74%. → New pooled rate: New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Point to each numbered stage and explain which recorded **input** it uses.

03. Fixed slice rates, changed pooled accuracy

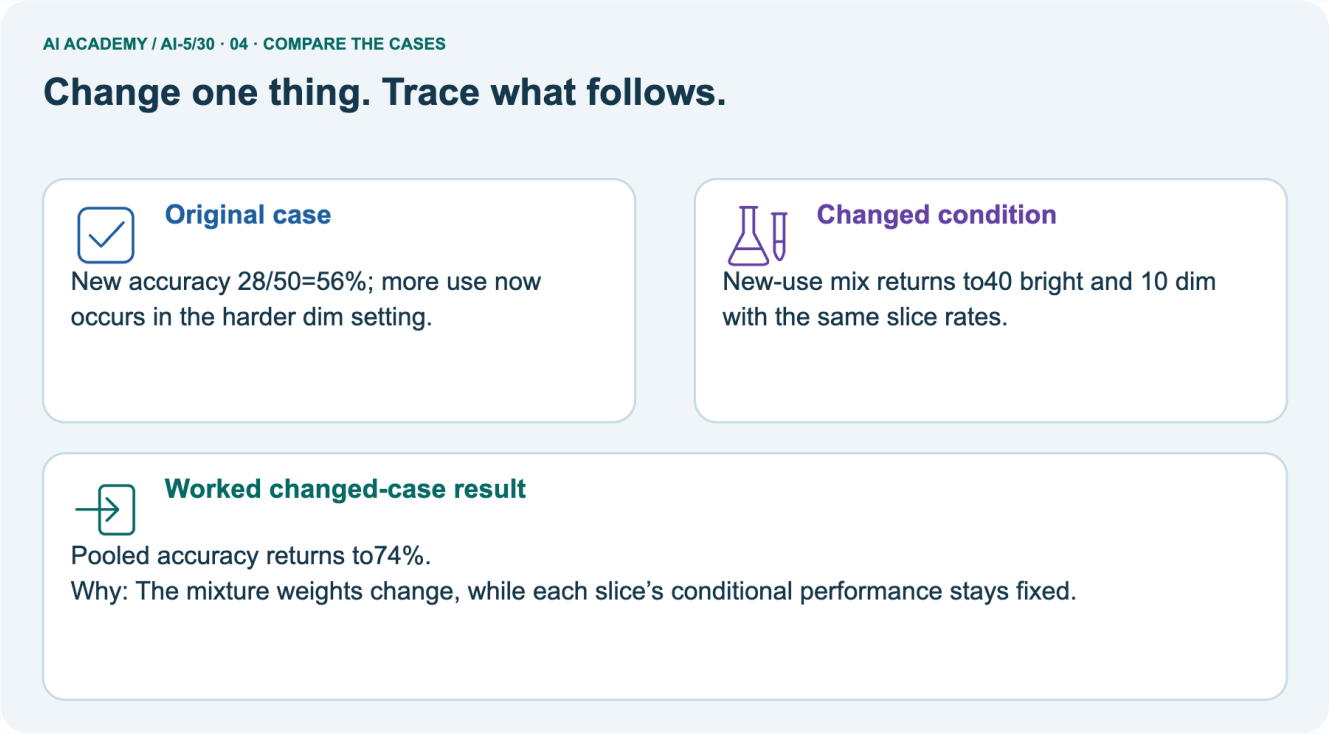


Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed slice rates, changed pooled accuracy. Units: accuracy percent. bright: 80. dim: 50. old pooled: 74. new pooled: 56.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



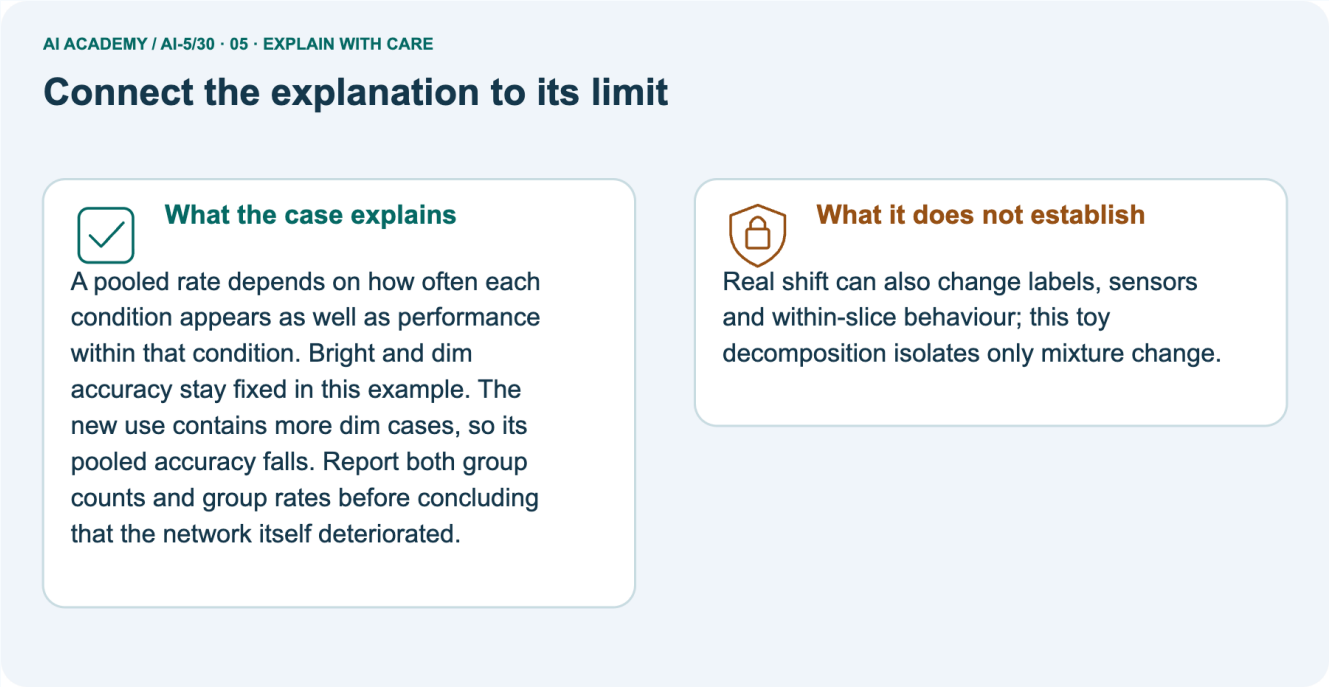
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New accuracy 28/50=56%; more use now occurs in the harder dim setting.
Changed condition: New-use mix returns to40 bright and 10 dim with the same slice rates.
Expected result: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

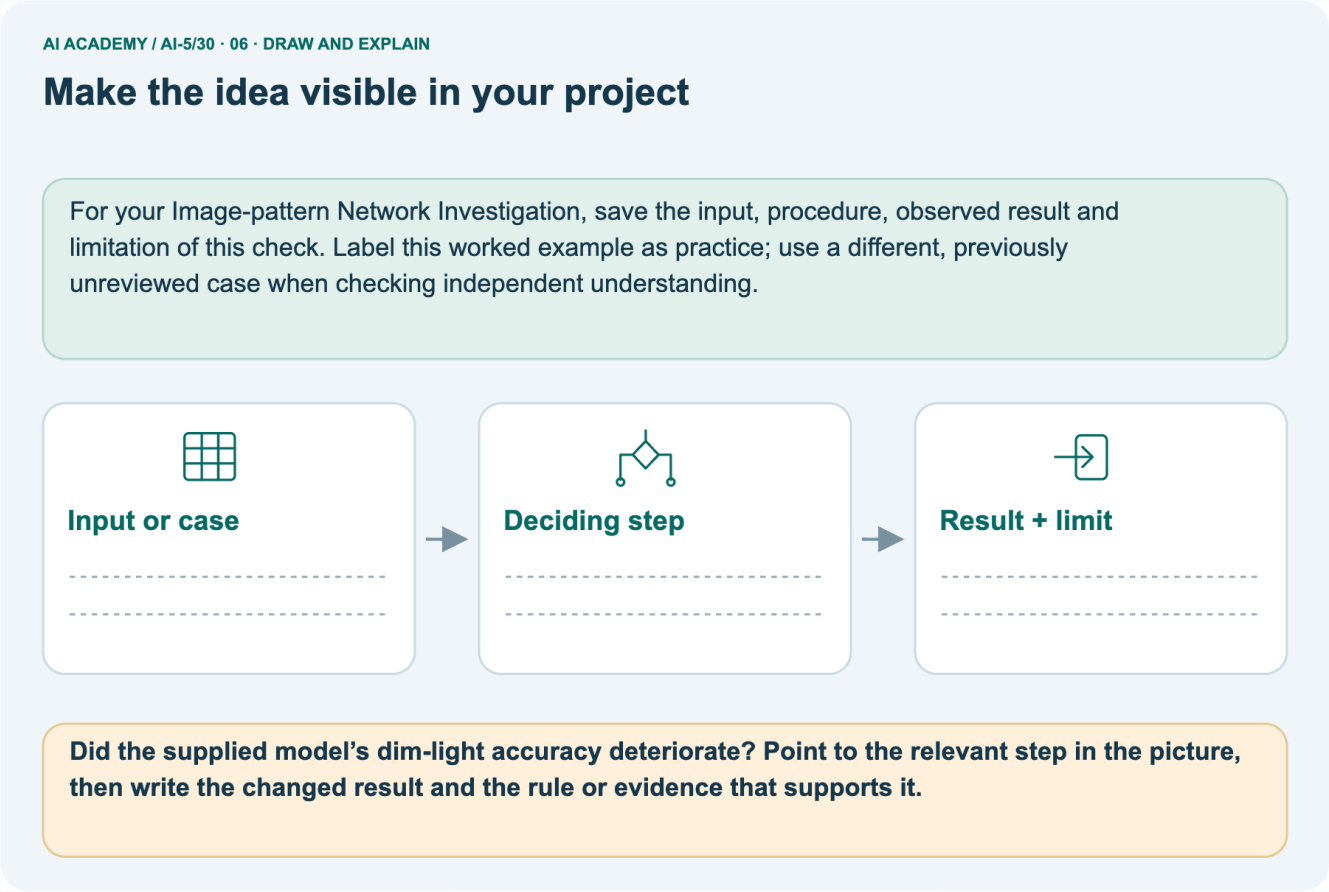
A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Limit: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Earlier wording, alternatives and picture transcripts

01. Dataset shift in neural models: the case

02. Follow the deciding steps

03. Fixed slice rates, changed pooled accuracy

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.